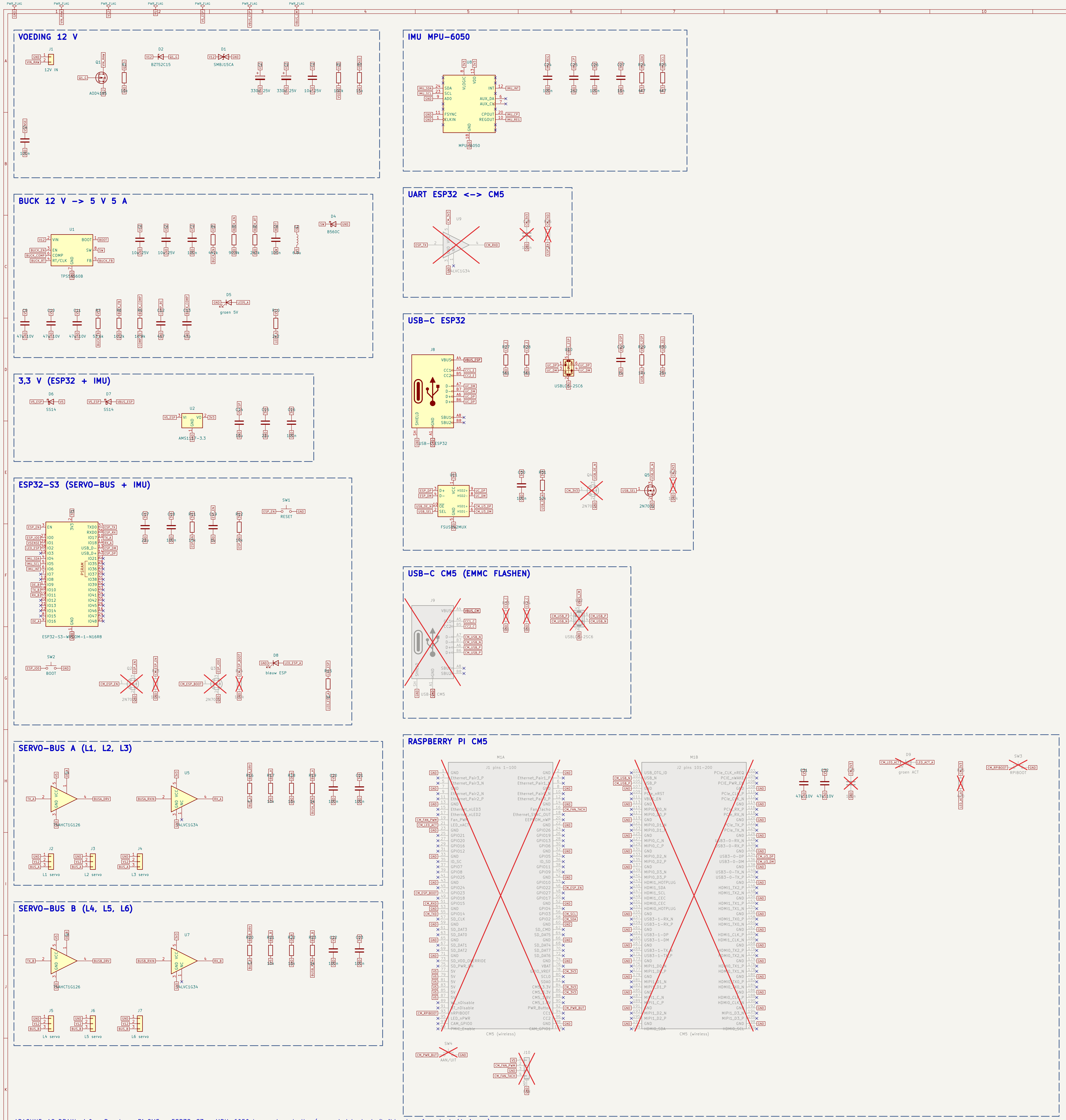

ARACHNE–18 BRAIN v1.0 – Raspberry Pi CM5 + ESP32–S3 + MPU–6050 hexapod controller (generated by tools/build_sch.py from tools/design.py)
Servo connectors: 1 = GND, 2 = +12 V, 3 = DATA (Feetech STS half–duplex bus, 1 Mbaud). CHECK the pin order against your cable before plugging in!
Bus A = legs L1 L2 L3 (ESP32 UART1: TX IO17, RX IO18, DE IO16). Bus B = legs L4 L5 L6 (UART2: TX IO10, RX IO11, DE IO9). DE = RTS in RS485 half–duplex mode.
CM5 is never back–powered: ESP–>CM5 UART through 74LVC1G34 powered by CM5_3V3, ESP reset/boot via 2N7002, USB switch disabled while the CM5 is off.
ESP32 USB: USB–C plugged in –> ESP32 talks to the PC; otherwise it is wired to CM5 USB3–0 (USB2 pair) –> /dev/ttyACM* on the Pi.
PI LINK header = pins 1–10 of a Raspberry Pi 40–pin header: a straight 10–way ribbon connects an external Pi 4 (5 V, GND, UART, GPIO4 = ESP reset) when no CM5 is fitted.
Buck = TI TPS54560B datasheet design (fig. 32): 400 kHz, 5.0 V 5 A. UVLO start 6.5 V.

CM5 + ESP32–S3 + MPU–6050, 6 servo–bus legs
Kobe Van Robaeys – maakleerplek
Sheet: /
File: arachne_brain.kicad_sch
Title: ARACHNE–18 brain board
Size: A1
Date: 2026–10–17
Rev: 1.0
KiCad E.D.A. 10.0.3
Id: 1/1